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BT 604 Current Final Terms Fall 2020

M.C.Q

1. The industrial Biotechnology is also known as _____. **White Biotechnology**

2. _____ are exploited due to their ability to ferment sugars to ethanol. **Yeasts**

3. L-form bacteria usually lack_____. **Cell wall**

4. E. Coli was discovered by_____. **Theodor escherich**

5. _____ relatively small circular extra chromosomal DNA molecules may also be present. **Plasmids**

5. Which statement is not true about the cell membrane in bacteria? **The outer side of cell membrane is hydrophilic**

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6. _____ element is essential for the synthesis of nucleic acids, intermediates of carbohydrate metabolism and compounds involved in energy transduction. **Phosphorus**

7. In _____ fresh medium are added as a single supplement and the volume of the batch increases with time. **Fed-batch systems**

8. Chlorine, hypochlorites, chlorine compounds and phenols are extensively used as_____. **Disinfectants**

9. Which is one of the following is also functions as a coenzyme for the synthesis of purine and pyrimidine bases of nucleic acids? **Folic acid**

10. (EMP) PATHWAY, is the most common route and found in all major group of organisms including _____ fungi, yeasts and many bacteria. **Filamentous**

11. Which one of the following is pathway produces only half the yield of atp compared with the emp pathway? **The phosphoketolase pathway**

12. _____ is produced by filamentous fungi, yeasts and some bacteria. **Trehalose**

13. _____is the biosynthesis of complex organic molecules from, simpler ones. **Anabolism**

14. The calvin cycle is essentially divided into three phases Carboxylation, Reduction and____. **Regeneration**

15. How many culture collections around the world? **500**

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16. Many of these restriction enzymes, found in various species of bacteria, make a staggered cut through a double-stranded DNA molecule at a specific sequence_____. **Palindrome**

17. Introduction of recombinant plasmids into bacterial cells can be achieved through _____treatment. **Calcium chloride**

18. Whey is an aqueous byproduct of the _____ industry. **Dairy**

19. When the water supply is 'hard', it is treated to remove salts such as_____. **Calcium carbonate**

20. Transfer into liquid from the gaseous phase is enhanced by_____. **Agitation**

21. The Environmental condition for the fermentation _____ condition. **Aseptic**

22. Liquid movement is initiated by the injection of ___at the bottom of the internal or external. **Compressed air**

23. Heat generated in the fermentation is primarily due to _____ of microorganisms. **Metabolic activity**

24. Which one of the following is used to maintain the pH of fermentation media? **Ammonium hydroxide**

25. _____ operation can extend the product formation phase and may overcome problems associate with the use of expressive, rapidly metabolized, substrates. **Fed-batch**

26. Steam may also be _____ directly into the head space above the fermentation medium. **Injected**

27. Enzymes, organic acids and ethanol are now produced by_____ particularly in areas where modern fermentation equipment is unavailable. **Solid substrate fermentations**

28. The rate of oxygen transfer is greatly influenced by the _____ of the substrate particles. **Size**

29. Broth conditioning techniques are mostly used in association with sedimentation and___. **Centrifugation**

30. Many recombinant proteins are formed as_____ which are insoluble inactive aggregates of over-expressed polypeptides. **Inclusion bodies**

31. In1959 as food and drug administration (FDA) classified MSG as 'generally regarded as safe_____. **(GRAS)**

32. Citric acid is widely used in the food industry as an acidulant and _____ agent in beverages. **Flavouring**

33. The term beer is given to _____alcoholic beverages made from partially germinated cereal grains, referred to as malt. **Non-distilled**

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Qno1: Which techniques are used in the fermentation broth? 2 marks

Turbidimetric and spectrophotometric methods require the construction of appropriate calibration curves, prepared using standard cell suspensions containing known concentrations of cells. Also, care must be taken when interpreting the results if the fermentation broth contains particulate matter or is highly coloured.

Qno2: Write down the applications of enzyme in different industries. 5 marks

Enzyme	Source	Use
α-Amylase EC 3.2.1.1	*Aspergillus* & *Bacillus* species	Digestive aids
Alcohol dehydrogenases EC 1.1.1.1 or alcohol oxidases EC 1.1.3.13	*Saccharomyces cerevisiae* *Candida boidinii* & *Pichia pastoris*	Ethanol testing
Asparaginase EC 3.5.1.1	*Escherichia coli*, *Serratia marcescens* & *Erwinia carotovora*	Maintains low asparagine levels in treating cancers (lymphomas and leukaemias) whose cells cannot synthesize the amino acid
Catalase EC 1.11.1.6	*Aspergillus niger*, *Corynebacterium glutamicum* & *Micrococcus lysodeikticus*	Contact lens cleaning systems
Cholesterol esterase EC 3.1.1.13	*Pseudomonas fluorescens*	Monitoring serum cholesterol levels
Creatininase EC 3.5.2.10 (creatinine amidohydrolase)	*Pseudomonas putida* & recombinant *Escherichia coli*	Determination of serum creatinine levels
Glucose oxidase EC 1.1.3.4	*Aspergillus niger*	Analysis of blood glucose levels
β-Lactamase EC 3.5.2.6	*Bacillus cereus* & *Escherichia coli*	Treatment of penicillin allergy
Proteases (various)	Several bacterial & fungal sources	Used as digestive aids, for wound debridement and contact lens cleaning
Rhodanase EC 2.8.1.1 (thiosulphate sulphur transferase)	*Trichoderma* species	Treatment for cyanide poisoning
Streptokinase EC 3.4.22.10 (a cysteine protease)	Various haemolytic streptocci	To break down blood clots
Urease EC 3.5.1.5	*Lactobacillus fermentum*	Removal of urea from blood in renal failure
Uricase EC 1.7.3.3 (urate oxidase)	*Arthrobacter globiformis* & *Candida utilis*	Used in the treatment and diagnosis of gout
Some microbial enzymes used in molecular biology (*DNA directed enzymes)		
DNA ligases (e.g. ATP, EC 6.5.1.1)	*Escherichia coli*	Joining pieces of DNA
DNA polymerase* EC 2.7.7.7	*Thermus aquaticus*	DNA synthesis, e.g. in PCR for forensic analysis
Restriction endonucleases e.g. *Bam* HI and *Eco* RI	*Bacillus amyloliquefaciens* H & *Escherichia coli* RY13	'Cutting' DNA at specific base sequences
RNA polymerase* EC 2.7.7.6	*Salmonella typhimurium*	RNA synthesis

Qno3: What are transposons or jumping gene? Why they are called jumping gene? 3 marks

Bacteriophages, like plasmids, May also acquire transposons, which are pieces of DNA. That can 'jump' from one piece of DNA to another, e.g. From a plasmid to a chromosome and vice versa. The bacteriophages can carry transposons on to new host bacterial cells, where they are able to 'jump' onto a plasmid or the host chromosomal DNA.

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Qno4: Write down the types of mutagens. 2 marks

Physical mutagens include ultraviolet, g and X radiation; and chemical mutagens are compounds such as ethane methane sulphonate (EMS), nitroso methyl guanidine (NTG), nitrous acid and acridine mustards.

Qno5: Explain the STRATEGIES FOR THE GENETIC ENGINEERING in detail. 5 marks

• Genetic engineering involves manipulation of DNA outside the cell.

• It necessitates the initial isolation and recovery of the gene(s) of interest from the donor organism's genome.

• Isolated DNA sequences may then be modified and the regulation of their expression altered, before insertion into host organisms via a suitable easily manipulated vector system.

• The first step requires total DNA extraction from the donor organism, which is then cut into smaller sequences using a specific restriction endonuclease.

• Many of these restriction enzymes, found in various species of bacteria, make a staggered cut through a double-stranded DNA molecule at a specific sequence or palindrome.

• As a result, the ends of cut molecules have complementary singlestranded sequences.

• The small sections of DNA (restriction fragments) can then be joined or spliced into vector DNA molecules that have been cut with the same restriction enzyme.

• Splicing is performed by an enzyme, DNA ligase, and creates a synthetic DNA molecule.

Qno6: Explain the processes which used to recover the cell harvesting. 10 marks

PRODUCT RECOVERY:

Following cell disruption, soluble proteins are usually separated from cell debris by centrifugation. The solubility of the salt varies with temperature, so strict temperature control is required. Many alkaloids, antibiotics, steroids and vitamins are recovered by liquid–liquid extraction methods using organic solvents. Chromatographic techniques are usually employed for higher value products. These methods, normally involving columns of chromatographic media (stationary phase), are used for desalting, concentration and purification of protein preparations. The order and choice of technique will depend on the particular product, but the following chromatographic parameters should be considered: capacity, recovery and resolving power (selectivity). Affinity

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chromatography is a particularly powerful and highly selective purification technique. Ligands are covalently linked to the matrix material, e.g. agarose, via a spacer arm to avoid steric hindrance. The loading capacity of affinity columns is large, as the volume of the sample is unimportant and high resolution can be attained For industrial-scale operations, it is often necessary to sterilize the chromatographic media in order to comply with various regulatory requirements. Gel filtration chromatography essentially involves separation on the basis of molecular size (molecular sieving), although molecular shape can also influence separation performance. The initial choice of stationary phase material is also a key factor, as some may interact with the target product, e.g. carbohydrate-based matrices interact with glycoprotein's. High-performance liquid chromatography (HPLC) was originally developed for the separation of organic molecules in nonaqueous solvents, but is now used for proteins in aqueous olution. The protein to be purified must have an affinity for this ion and binds to it by forming coordination complexes with groups such as the imidazole of histidine residues. Membrane separation techniques are primarily used for the removal of low molecular weight solutes and inorganic ions from a solution.

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